

Integrated Digital Timing System

A Unified Verilog HDL Design Integrating
Digital Clock, Stopwatch, and Alarm Modules
with a Common Seven-Segment Display Controller

Project Team

Priya Nageswari Karanam

Satish Chitikela

Devi Veeti

Arvind Pandhi

Abstract

This project presents the design, integration, and functional verification of an **Integrated Digital Timing System** using **Verilog HDL**. The system integrates three independent timing modules—a **24-hour Digital Clock**, **Stopwatch**, and **Alarm**—into a unified hardware architecture with a common control interface. A multiplexed **four-digit seven-segment display** is used to display the output of the selected module, providing efficient hardware utilization and a user-friendly interface.

The Digital Clock supports manual time loading and continuous timekeeping in a 24-hour format. The Stopwatch provides start, stop, and reset operations with debounced push-button inputs for reliable operation. The Alarm module enables programmable alarm settings and includes snooze and stop functionalities with an integrated buzzer for audible notification.

The system is designed using a modular RTL architecture, allowing each subsystem to be developed, verified, and integrated independently. The integrated design was functionally verified through simulation, confirming the correct operation of all modules individually and as a unified system. This project demonstrates the practical implementation of modular digital design principles and the successful integration of multiple timing functionalities into a single hardware system.

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1 Introduction

Time plays an important role in almost every electronic system. Whether it is displaying the current time, measuring the duration of an event, or notifying a user at a specific moment, timing functions are essential in many real-world applications. These functions are commonly found in devices such as digital watches, home appliances, industrial controllers, and embedded systems.

In digital hardware design, developing such systems provides an excellent opportunity to understand concepts such as sequential logic, finite state machines, counters, control signals, and modular design. Building multiple timing functions within a single hardware platform also demonstrates how independent modules can interact while sharing common resources efficiently.

This project applies these digital design concepts to develop an integrated timing solution using **Verilog HDL**. It emphasizes modular RTL design, subsystem integration, and functional verification through simulation, providing practical experience in designing, integrating, and validating complex digital systems.

2 Project Overview

This section presents the objectives and technical specifications of the proposed Integrated Digital Timing System, providing a high-level overview of its design and functionality.

2.1 Objectives

- Design a 24-hour digital clock using Verilog HDL.
- Implement a stopwatch with start, stop, and reset functionalities.
- Develop a programmable alarm with snooze and stop features.
- Integrate all modules into a unified hardware architecture.
- Display the selected module output using a multiplexed four-digit seven-segment display.
- Functionally verify the complete design through simulation.

3 System Architecture

3.1 Overall Block Diagram

Figure 1 illustrates the overall architecture of the proposed Integrated Digital Timing System. The design integrates three independent timing modules—a Digital Clock, Stopwatch, and Alarm—under a common top-level integration module. User inputs are processed through dedicated control signals, while a debounce circuit ensures reliable push-button operation. Based on the selected operating mode, the top-level module enables the appropriate subsystem and routes its output through the mode selection logic. The selected timing information is then displayed on a shared four-digit seven-segment display using a multiplexed display controller. The Alarm module additionally interfaces with a buzzer driver to generate an audible alert when the programmed alarm condition is satisfied.

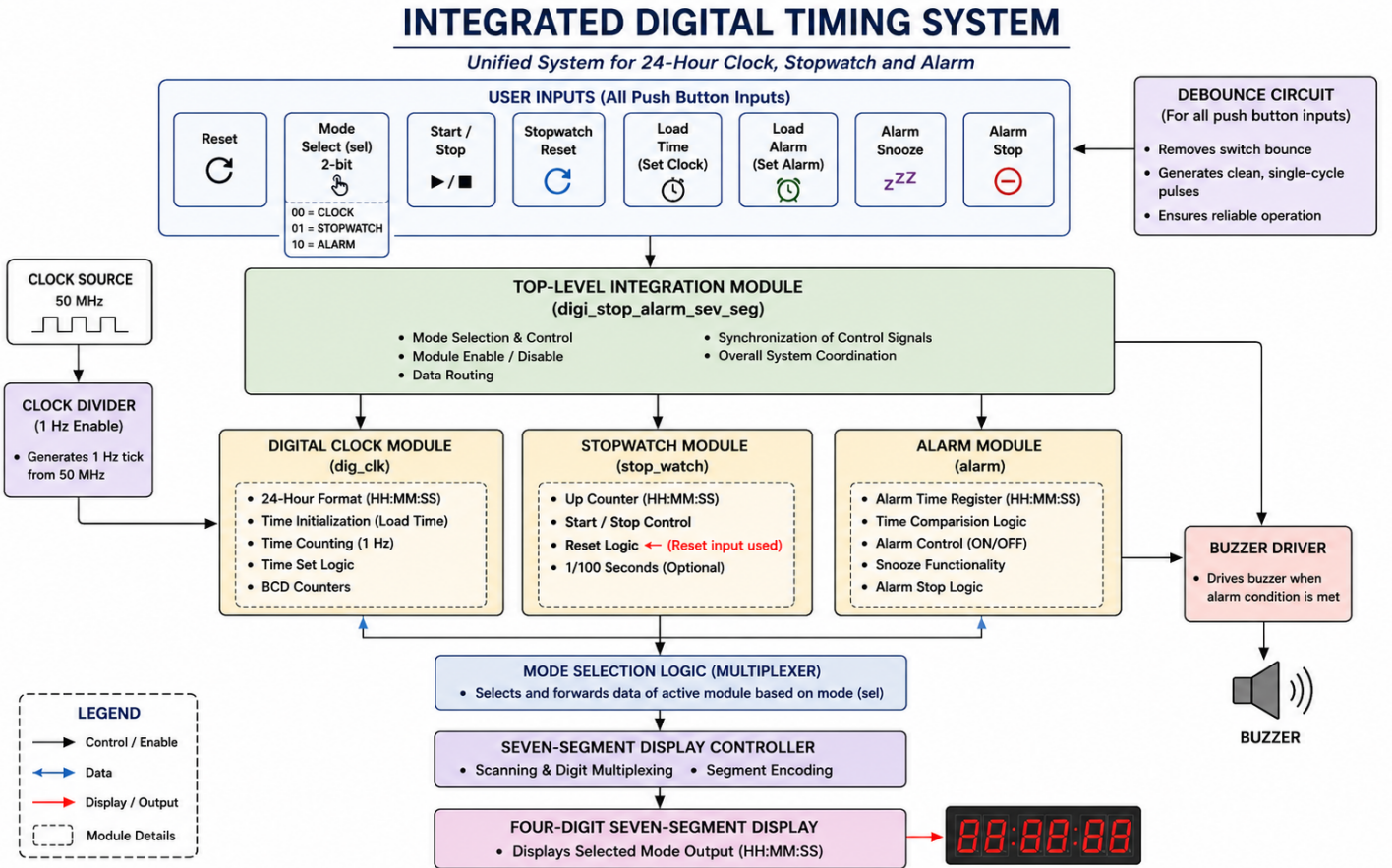


Figure 1: Overall Block Diagram of the Integrated Digital Timing System

3.2 System Operation

The overall operation of the Integrated Digital Timing System begins with user inputs such as mode selection, reset, start/stop control, time loading, alarm loading, snooze, and alarm stop. All push-button inputs are processed through a debounce circuit to eliminate switch bouncing and ensure reliable operation.

The top-level integration module receives these control signals and coordinates the operation of the Digital Clock, Stopwatch, and Alarm modules. Depending on the selected operating mode (`sel`), the corresponding module is enabled while the remaining modules continue to operate independently in the background where required.

The output of the selected module is forwarded to the mode selection logic (multiplexer), which routes the appropriate timing information to the seven-segment display controller. The display controller performs multiplexing and segment encoding to drive the shared four-digit seven-segment display.

The Alarm module continuously compares the current clock time with the user-programmed alarm time. When both values match, the alarm condition is activated and the buzzer driver generates an audible alert. The alarm can subsequently be snoozed or stopped using the dedicated control inputs.

4 Design Implementation

This section presents the implementation of the functional modules that constitute the Integrated Digital Timing System. Each module is developed using a modular RTL design approach and integrated to achieve the complete system functionality. The following sections briefly describe the operation of each module.

4.1 Digital Clock

The Digital Clock module is responsible for maintaining and updating the current time in a 24-hour format. It is designed using cascaded Binary Coded Decimal (BCD) counters, where each counter represents a single digit of the displayed time. The least significant minute digit is incremented with every clock pulse, and whenever a digit reaches its maximum value, a carry is generated to increment the next higher digit. This cascading mechanism ensures accurate time progression while maintaining valid BCD values. The module also provides a manual time loading feature, enabling the user to initialize the clock to any desired time before normal operation begins. Once the time reaches 23:59, the clock automatically rolls over to 00:00

and continues counting. The generated time values are continuously provided to the display controller for visualization and to the Alarm module for alarm time comparison.

4.2 Stopwatch

The Stopwatch module is designed to measure elapsed time independently of the Digital Clock. Unlike the clock, it operates only when enabled by the user through the start control signal and can be paused using the stop control signal. To ensure reliable operation, both control inputs are passed through debounce circuits before enabling or disabling the stopwatch. The timing mechanism is implemented using cascaded BCD counters to measure elapsed time while maintaining valid BCD values throughout the counting process. The stopwatch operates independently of the Digital Clock and Alarm modules, allowing elapsed time to be measured without affecting the operation of the remaining subsystems. The current stopwatch value is continuously available to the display controller whenever the stopwatch mode is selected.

4.3 Alarm

The Alarm module is responsible for notifying the user when the current time matches the programmed alarm time. It continuously compares the output of the Digital Clock with the stored alarm value and activates the buzzer whenever both values are identical. The alarm time can be programmed by the user through dedicated load inputs, allowing flexible alarm configuration. To improve usability, the module supports both snooze and stop functionalities. When the snooze button is pressed, the alarm is temporarily postponed by five minutes, while the stop button immediately disables the buzzer and clears the current alarm condition. By continuously monitoring the clock output and responding to user controls, the Alarm module provides a reliable and user-friendly alert mechanism within the integrated timing system.

4.4 Debounce Circuit

Mechanical push buttons often produce multiple unwanted transitions, known as switch bouncing, when pressed or released. These false transitions can cause multiple detections of a single button press, leading to incorrect system behavior. The Debounce Circuit eliminates this problem by monitoring the input signal for a predefined number of clock cycles before accepting it as a valid transition. In this project, the debounce circuit is used for the **start** and **stop** inputs of the Stopwatch module, ensuring that each

button press is detected only once and providing reliable user interaction.

4.5 Buzzer

The Buzzer module provides an audible indication whenever the programmed alarm time is reached. It receives the trigger signal from the Alarm module and generates an oscillating output by continuously toggling the buzzer signal. This periodic switching causes the internal vibrating element of the buzzer to oscillate, producing the audible beep. The buzzer remains active for a predefined duration and can be controlled using the `snooze` and `alarm_stop` inputs. The `snooze` function temporarily suppresses the buzzer while postponing the alarm, whereas the `alarm_stop` function immediately disables the buzzer and resets its internal state. By separating the buzzer control from the Alarm module, the design maintains a modular and organized system architecture.

4.6 Seven-Segment Display Controller

The Seven-Segment Display Controller is responsible for displaying the output of the selected timing module on a shared four-digit seven-segment display. It receives the hour and minute digits from the top-level integration module and converts each Binary Coded Decimal (BCD) value into its corresponding seven-segment code using a decoder. A multiplexing technique is employed to activate one display digit at a time at a high switching frequency, creating the illusion that all four digits are illuminated simultaneously. This approach minimizes hardware requirements while providing a stable and flicker-free display. The controller continuously updates the displayed values based on the selected operating mode, ensuring that the Digital Clock, Stopwatch, or Alarm time is presented correctly to the user.

4.7 Top-Level Integration Module

The Top-Level Integration Module serves as the central controller of the Integrated Digital Timing System. It interconnects the Digital Clock, Stopwatch, Alarm, and Seven-Segment Display Controller into a single unified architecture. Based on the mode selection input (`sel`), the module selects the appropriate timing information from the Digital Clock, Stopwatch, or Alarm module and forwards it to the display controller. It also manages the loading of alarm values, routes the current clock time to the Alarm module for continuous comparison, and coordinates user control signals such as reset, start, stop, snooze, and alarm

stop. By integrating all functional modules through a common interface, the design ensures synchronized operation while maintaining a modular and scalable architecture.

5 Functional Verification

The functionality of the Integrated Digital Timing System was verified using a Verilog testbench. Individual modules were first tested independently and then integrated into the complete system. Various input conditions were applied to validate the correct operation of the Digital Clock, Stopwatch, Alarm, Debounce Circuit, Buzzer, and Seven-Segment Display Controller. The observed simulation waveforms confirmed that the system operated as expected under different operating conditions.

5.1 Test Scenarios

Table 1: Functional Verification Test Scenarios

Test Case	Description	Result
TC1	Verify system initialization after reset.	Passed
TC2	Verify manual loading of the Digital Clock time.	Passed
TC3	Verify continuous 24-hour clock operation.	Passed
TC4	Verify Stopwatch start, stop, and reset functionalities.	Passed
TC5	Verify loading of the programmed alarm time.	Passed
TC6	Verify alarm activation and buzzer operation when the current time matches the alarm time.	Passed
TC7	Verify snooze functionality by postponing the alarm for five minutes.	Passed
TC8	Verify alarm stop functionality and buzzer deactivation.	Passed
TC9	Verify mode selection and correct display switching between Digital Clock, Stopwatch, and Alarm.	Passed

6 Results

This chapter presents the final outputs obtained from the integrated digital timing system. The RTL diagram illustrates the complete hardware architecture after module integration, while the simulation waveform verifies the correct operation of the system under various operating conditions.

6.1 RTL Diagram

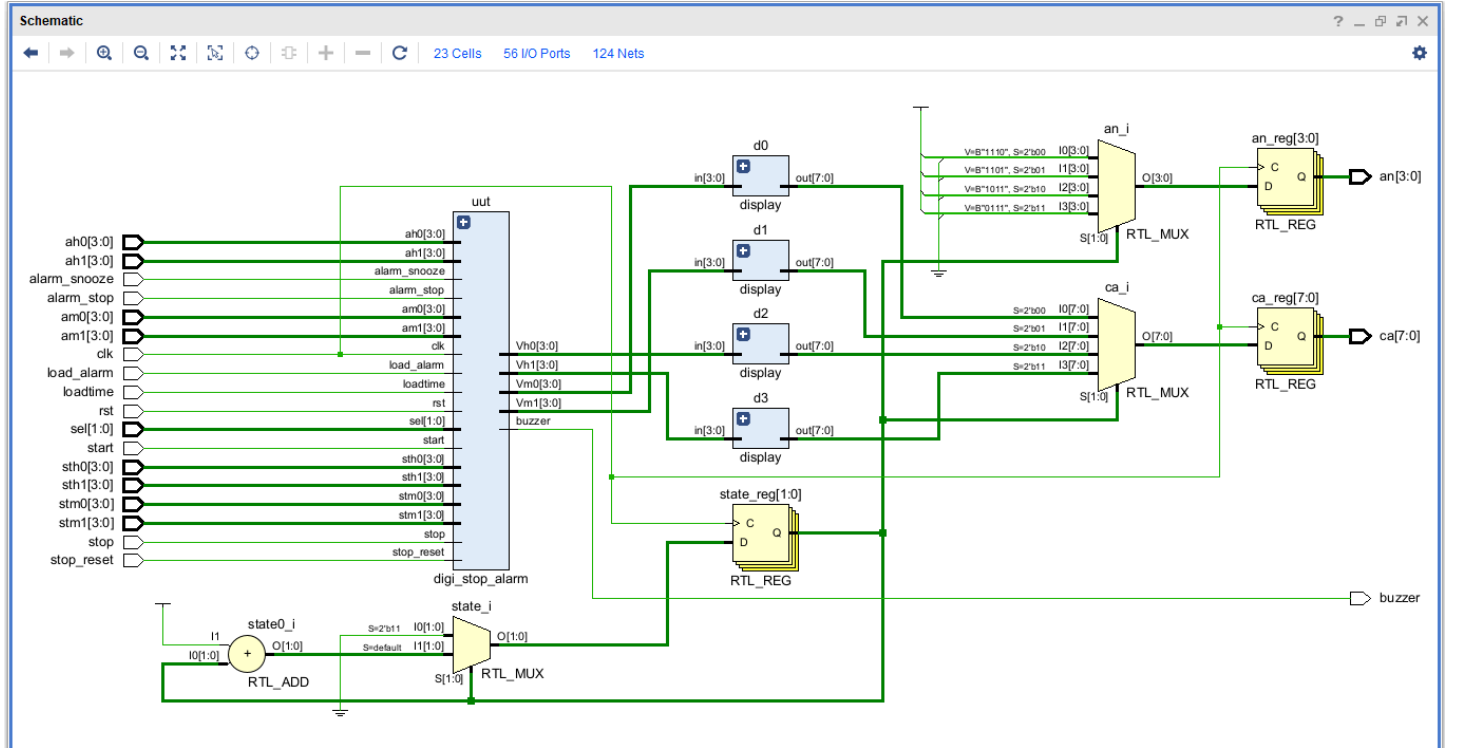


Figure 2: RTL Diagram of the Integrated Digital Timing System

6.2 Simulation Results

The integrated design was simulated to verify the functional behavior of the Digital Clock, Stopwatch, Alarm, and display interface. The following waveforms demonstrate the response of the system to different control inputs and confirm the correct operation of the integrated modules.

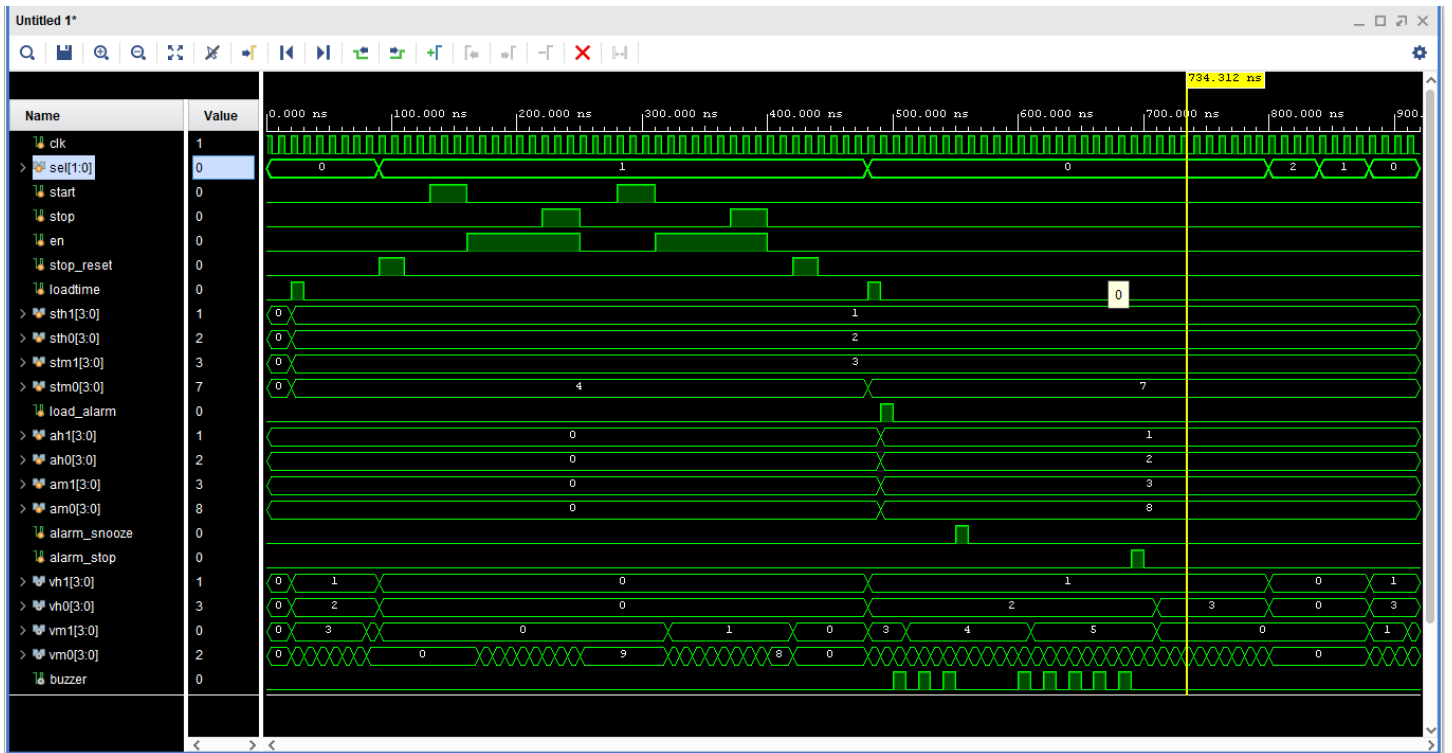


Figure 3: Simulation Waveform of Integrated System Operation

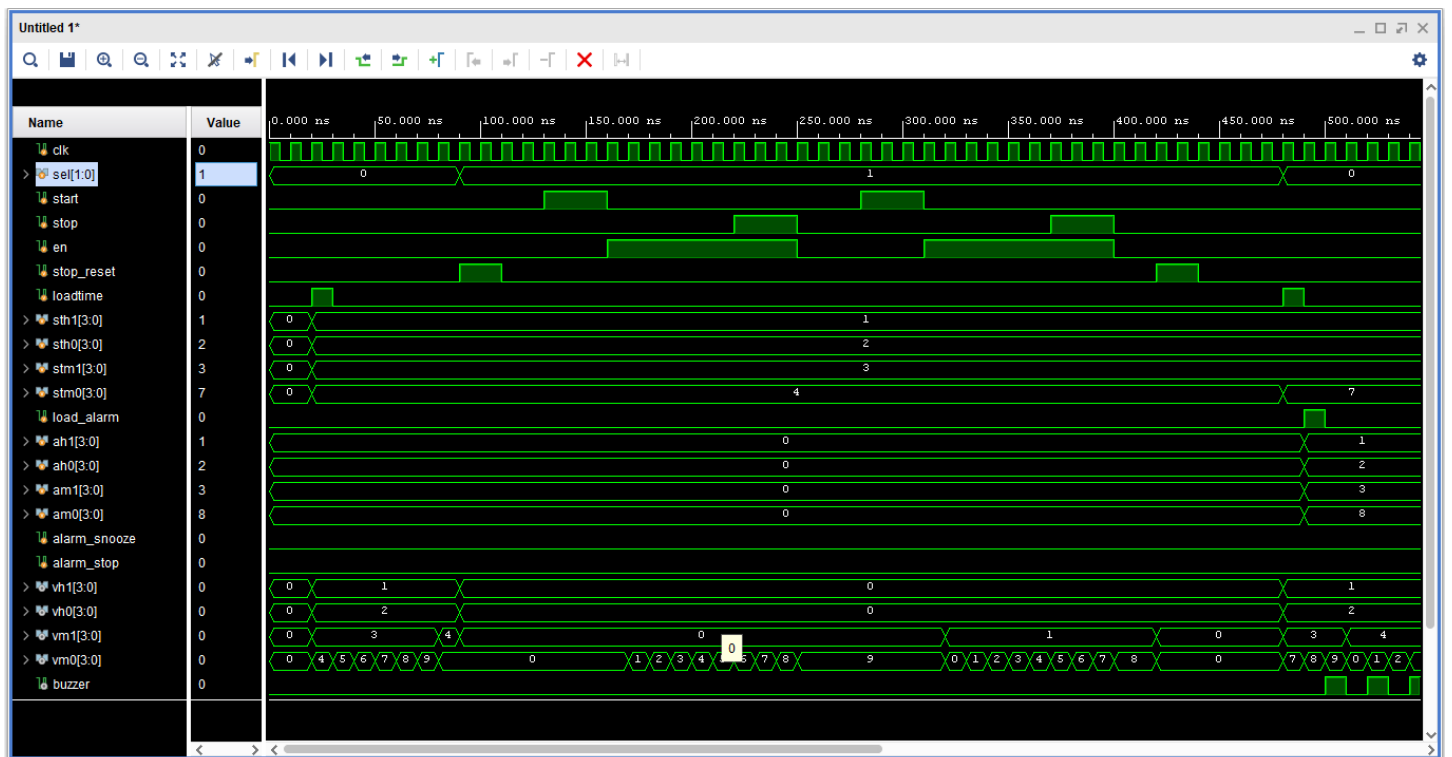


Figure 4: Simulation Waveform of Digital Clock and Stopwatch Operations

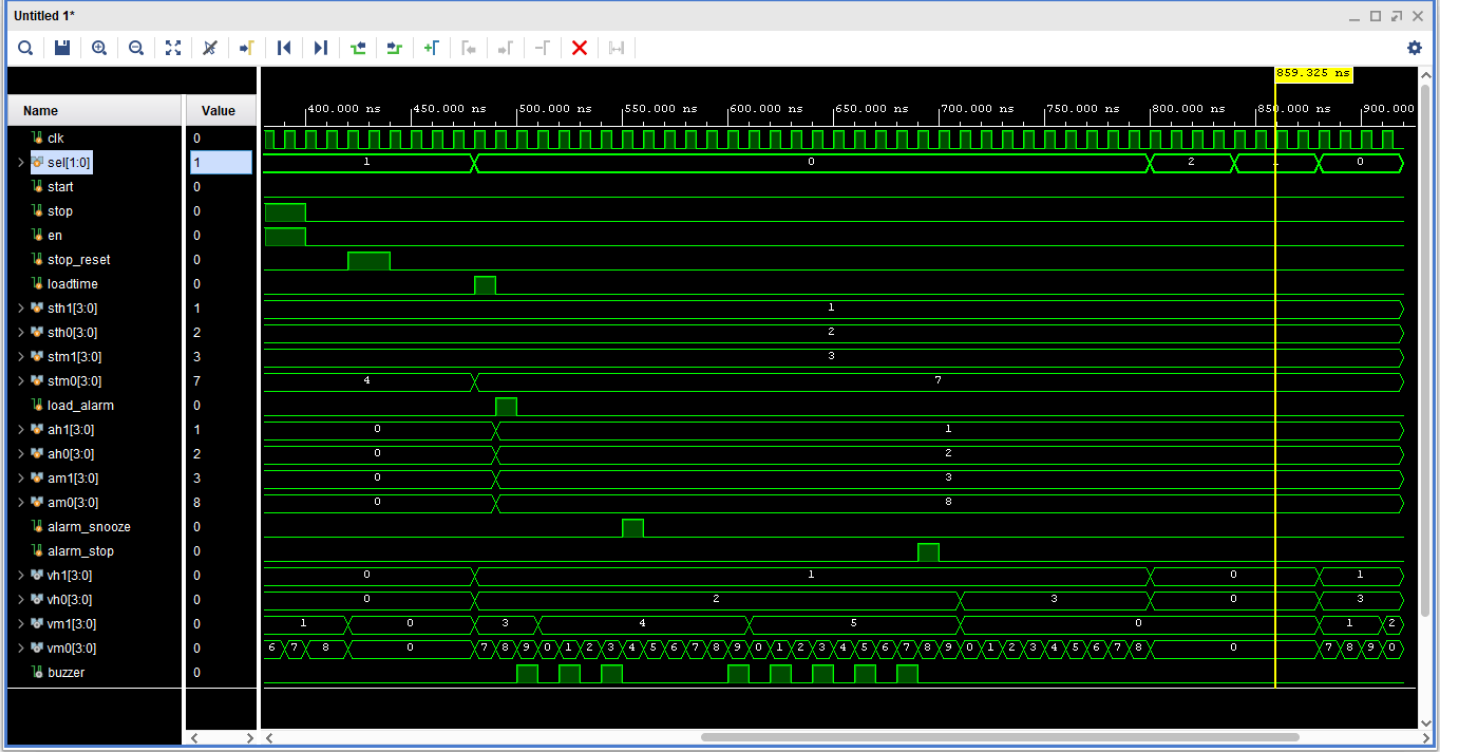


Figure 5: Simulation Waveform of Alarm and Buzzer Operations

7 Applications

The integrated timing architecture developed in this project can serve as a foundation for a variety of digital timing and monitoring systems. The combination of timekeeping, elapsed-time measurement, programmable alerts, and multiplexed display control can be adapted to applications such as:

- **Digital Timekeeping:** Basic electronic clocks and time-display units.
- **Event Timing:** Stopwatch-based systems for measuring the duration of events and processes.
- **Alert Systems:** Programmable alarm units for reminders and scheduled notifications.
- **Embedded Interfaces:** Timing and display subsystems within embedded digital controllers.
- **FPGA-Based Systems:** A modular reference design for implementing and extending timing functions on FPGA platforms.

8 Future Trends

The Integrated Digital Timing System can be improved with more features in the future. Some possible improvements are:

- **Accurate Timekeeping:** Add a real-time clock (RTC) for accurate timekeeping.
- **Multiple Alarms:** Allow the user to set more than one alarm.
- **Better User Interface:** Add a keypad or touch interface for easier control.
- **More Display Options:** Add date, day, and other useful information to the display.
- **Wireless Control:** Add Bluetooth or Wi-Fi to control and monitor the system remotely.
- **FPGA Implementation:** Implement the complete design on an FPGA board and test it in real time.

9 Conclusion

The Integrated Digital Timing System was successfully designed using Verilog HDL by combining a Digital Clock, Stopwatch, Alarm, Buzzer, and Seven-Segment Display Controller into a single system. The modular design approach made it possible to develop and verify each function independently before integrating the complete system. Functional simulation confirmed the correct operation of time loading, clock counting, stopwatch control, alarm triggering, snooze, alarm stop, and display selection. The project provided practical understanding of RTL design, module integration, control signals, and functional verification, forming a strong foundation for further development of digital and FPGA-based systems.